



Week 7 - Graded Assignment 7



The due date for submitting this assignment has passed.

Due on 2026-04-01, 23:59 IST.

You may submit any number of times before the due date. The final submission will be considered for grading.

You have last submitted on: 2026-04-01, 07:55 IST

Note: This assignment will be evaluated after the deadline passes. You will get your score 48 hrs after the deadline. Until then the score will be shown as Zero.

1 point

Consider the vector spaces V and functions $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$ defined as follows:

i) $V = \mathbb{R}^2$ and $\langle (x_1, x_2), (y_1, y_2) \rangle = x_1 y_1 - x_2 y_1 - x_1 y_2 + 2x_2 y_2$. $\langle (x_1, x_2), (y_1, y_2) \rangle = x_1 y_1 - x_2 y_1 - x_1 y_2 + 2x_2 y_2$.

ii) $V = M_{2 \times 2}(\mathbb{R})$ and $\langle A, B \rangle = \text{Tr}(AB)$, $\langle A, B \rangle = \text{Tr}(AB)$, where $\text{Tr}(M)$ denotes the trace of a matrix M , i.e., the sum of the diagonal elements of M .

iii) $V = M_{2 \times 1}(\mathbb{R})$ and $\langle A, B \rangle = \text{Tr}(AB^t)$, $\langle A, B \rangle = \text{Tr}(AB^t)$, where $\text{Tr}(X)$ denotes the trace of a matrix X , i.e., the sum of the diagonal elements of X . Y^t denotes the transpose of matrix Y .

iv) $V = \mathbb{R}^2$ and $\langle (x_1, x_2), (y_1, y_2) \rangle = x_1 y_2 + x_2 y_1$. $\langle (x_1, x_2), (y_1, y_2) \rangle = x_1 y_2 + x_2 y_1$.

- ☒ (i) is an inner product.
- ☒ (ii) is an inner product.
- ☐ (iii) is an inner product.
- ☐ (iv) is an inner product.

No, the answer is incorrect.

Score: 0

Accepted Answers:

- (i) is an inner product.
- (iii) is an inner product.

1 point

Consider two linear transformations T and S from $\mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined as $T(x, y) = (2x + y, x + y)$ and $S(x, y) = (x + cy, x + 2y)$. Let A and B be matrix representations of linear transformations T and S with respect to the standard bases of $\mathbb{R}^2$ respectively. Consider the following statements:

- P:** If $c = 1$, then A and B are similar matrices.
- Q:** If $c = 2$, then A and B are similar matrices.
- R:** If $c = 1$ and $P^{-1}AP = B$, then P can be the matrix $\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$.
- S:** If $c = 1$ and $P^{-1}AP = B$, then P can be the matrix $\begin{bmatrix} 1 & 2 \\ 1 & 1 \end{bmatrix}$.
- T:** If $c = 1$, then there are infinitely many P satisfying the equation $P^{-1}AP = B$.

Which of the following options are true?

- ☒ P is true but Q is false.
- ☐



Both P and QP and Q are true.

☐

Both R and SR and S are true.

☒

RR is false but SS is true.

☒

TT is true.

Yes, the answer is correct.

Score: 1

Accepted Answers:

PP is true but QQ is false.

RR is false but SS is true.

TT is true.

1 point

Let $\langle \cdot, \cdot \rangle$ denote the standard inner product on $\mathbb{R}^2$, i.e., $\langle (x_1, x_2), (y_1, y_2) \rangle = x_1 y_1 + x_2 y_2$. Which one of the following options is (are) true for the vector $\gamma \in \mathbb{R}^2$, such that $\langle \alpha, \gamma \rangle = 4$ and $\langle \beta, \gamma \rangle = 8$, where $\alpha = (3, 1)$ and $\beta = (6, 2)$.

☐

No such γ exists.

☒

There are infinitely many such vectors which satisfy the properties of γ .

☐

γ is unique in $\mathbb{R}^2$.

☒

Any vector in the set $\{(t, 4 - 3t) \mid t \in \mathbb{R}\}$ satisfies the properties of γ .

☐

$(1, 1)$ is the only vector which satisfies the properties of γ .

Yes, the answer is correct.

Score: 1

Accepted Answers:

There are infinitely many such vectors which satisfy the properties of γ .

Any vector in the set $\{(t, 4 - 3t) \mid t \in \mathbb{R}\}$ satisfies the properties of γ .

1 point

Let $\langle \cdot, \cdot \rangle$ denote the standard inner product on $\mathbb{R}^2$, i.e., $\langle (x_1, x_2), (y_1, y_2) \rangle = x_1 y_1 + x_2 y_2$ and let $v \in \mathbb{R}^2$. Consider a linear transformation $T_v : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined as:

$$T_v(u) = \langle u, v \rangle v,$$

where $v \in \mathbb{R}^2$. Which of the following options is (are) true for T_v ?

☐

T_v is one-one for all $v \neq 0$.

☒

T_v is onto for all $v \neq 0$.

☐

T_v is onto for all $v \in \mathbb{R}^2$.

☒

T_v is not one-one for every $v \in \mathbb{R}^2$.

☒

There exists a $v \in R^2$ $v \in R^2$ such that T_v T_v is not onto.

Yes, the answer is correct.

Score: 1

Accepted Answers:

T_v T_v is onto for all $v \neq 0 \in R^2$ $v \neq 0 \in R^2$.

T_v T_v is not one-one for every $v \in R^2$ $v \in R^2$.

There exists a $v \in R^2$ $v \in R^2$ such that T_v T_v is not onto.

1 point

Let A and B be square matrices of the same order n . Which of the following statements are sufficient to conclude that A is equivalent to B.

☐ $\det(A) = \det(B)$.

☒ $\text{Nullity}(A) = \text{Nullity}(B)$.

☐

The set of columns of AA, and the set of columns of BB are both linearly dependent subsets of R^n R^n .

☒

There exists a matrix CC such that AA is equivalent to CC, and BB is equivalent to CC.

☒

There exist invertible matrices PP and QQ of order n such that $PA = BQ$ $PA = BQ$.

Yes, the answer is correct.

Score: 1

Accepted Answers:

$\text{Nullity}(A) = \text{Nullity}(B)$.

There exists a matrix CC such that AA is equivalent to CC, and BB is equivalent to CC.

There exist invertible matrices PP and QQ of order n such that $PA = BQ$ $PA = BQ$.

1 point

Let LL and L' L' be affine subspaces of R^3 R^3 , where $L = (0, 1, 1) + U$ $L = (0, 1, 1) + U$ and

$L' = (0, 1, 0) + U'$ $L' = (0, 1, 0) + U'$, for some vector subspaces UU and U' U' of R^3 R^3 . Let a basis

for UU be given by $\{(1, 1, 0), (1, 0, 1)\}$ $\{(1, 1, 0), (1, 0, 1)\}$ and a basis for U' U' be given by $\{(1, 0, 0)\}$

$\{(1, 0, 0)\}$. Suppose there is a linear transformation $T: U \rightarrow U'$ $T: U \rightarrow U'$ such that $(1, 0, 1) \in \ker(T)$

$(1, 0, 1) \in \ker(T)$ and $T(1, 1, 0) = (1, 0, 0)$ $T(1, 1, 0) = (1, 0, 0)$. An affine mapping $f: L \rightarrow L'$

$f: L \rightarrow L'$ is obtained by defining $f((0, 1, 1) + u) = (0, 1, 0) + T(u)$ $f((0, 1, 1) + u) = (0, 1, 0) + T(u)$, for

all $u \in U$ $u \in U$. Which of the following options are true?

☒

$L = \{(x, y + 1, x - y + 1) \mid x, y \in R\}$ $L = \{(x, y + 1, x - y + 1) \mid x, y \in R\}$.

☐

$L' = \{(x, y + 1, 0) \mid x, y \in R\}$ $L' = \{(x, y + 1, 0) \mid x, y \in R\}$.

☐

$L = \{(x - y, y + 1, x - y + 1) \mid x, y \in R\}$ $L = \{(x - y, y + 1, x - y + 1) \mid x, y \in R\}$.

☐

$L = \{(x, x + 1, y + 1) \mid x, y \in R\}$ $L = \{(x, x + 1, y + 1) \mid x, y \in R\}$.

☒

$f(x, y + 1, x - y + 1) = (y, 1, 0)$ $f(x, y + 1, x - y + 1) = (y, 1, 0)$

☐

$f(x - y, y + 1, x - y + 1) = (x, y + 1, 0)$ $f(x - y, y + 1, x - y + 1) = (x, y + 1, 0)$

☐

$f(x, x + 1, y + 1) = (y, 1, 0)$ $f(x, x + 1, y + 1) = (y, 1, 0)$

☐

$f(x, y + 1, x - y + 1) = (0, 1, y)$ $f(x, y + 1, x - y + 1) = (0, 1, y)$

Yes, the answer is correct.

Score: 1

Accepted Answers:

$$L = \{(x, y + 1, x - y + 1) \mid x, y \in \mathbb{R}\} \quad L = \{(x, y + 1, x - y + 1) \mid x, y \in \mathbb{R}\}.$$

$$f(x, y + 1, x - y + 1) = (y, 1, 0) \quad f(x, y + 1, x - y + 1) = (y, 1, 0)$$

Let θ be the angle between the vectors $u = (4, 7, 3)$ and $v = (1, 2, -6)$, then what will be the value of $\cos(\theta)$?

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 0

1 point

Consider a basis

$$\{v_1 = (1, 2, 0), v_2 = (2, -1, 0), v_3 = (0, 0, 2)\} \quad \{v_1 = (1, 2, 0), v_2 = (2, -1, 0), v_3 = (0, 0, 2)\}$$

of $\mathbb{R}^3$ with usual inner product. Suppose $v = (x, y, \frac{3x+y}{5}) \in V$ is written as $v = c_1 v_1 + c_2 v_2 + c_3 v_3$, such that $c_1 + c_2 = 4$. What will be the value of c_3 ?

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 2

1 point

Let $V = \mathbb{R}^2$ be a vector space. Consider two inner products $\langle \cdot, \cdot \rangle_1$ and $\langle \cdot, \cdot \rangle_2$ on V defined as

$$\langle (x_1, y_1), (x_2, y_2) \rangle_1 = x_1 x_2 - x_1 y_2 - x_2 y_1 + 4y_1 y_2 \quad \langle (x_1, y_1), (x_2, y_2) \rangle_1 = x_1 x_2 - x_1 y_2 - x_2 y_1 + 4y_1 y_2$$

and

$$\langle (x_1, y_1), (x_2, y_2) \rangle_2 = 3x_1 x_2 + 2y_1 y_2 \quad \langle (x_1, y_1), (x_2, y_2) \rangle_2 = 3x_1 x_2 + 2y_1 y_2.$$

If $\langle (a, b), (4, 3) \rangle_1 = 35$ and $\langle (a, b), (4, 3) \rangle_2 = 60$, then find the value $a + 2b$.

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 11

1 point

Let $V = \mathbb{R}^3$ be the inner product space with usual inner product. If θ is the angle between $(15, 8, 18)$ and (a, b, c) where $15a + 8b + 18c = 0$ and $(a^2 + b^2 + c^2) \neq 0$, then find the value of $\cos \theta$.

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 0

1 point

Consider a vector $\mathbf{v} = (x - 20, 2, 1) \in \mathbb{R}^3$. Find the value of x so that the length of the vector $\mathbf{v}$ is minimum.

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) 20

1 point

Consider the following subsets of $\mathbb{R}^3$, with $\langle \cdot, \cdot \rangle$ used to denote the standard dot product on $\mathbb{R}^3$, and $\|\cdot\|$ used to denote the standard length function.

$S_1 = \{\mathbf{v} \in \mathbb{R}^3 \mid \langle \mathbf{v}, (0, 0, 1) \rangle = 2, \|\mathbf{v}\| = 1\}$
 $S_2 = \{\mathbf{u} \in \mathbb{R}^3 \mid \mathbf{u} \in \text{Span}(\{(1, 1, 1)\}), \|\mathbf{u}\| = 1\}$
 $S_1 = \{\mathbf{v} \in \mathbb{R}^3 \mid \langle \mathbf{v}, (0, 0, 1) \rangle = 2, \|\mathbf{v}\| = 1\}$
 $S_2 = \{\mathbf{u} \in \mathbb{R}^3 \mid \mathbf{u} \in \text{Span}(\{(1, 1, 1)\}), \|\mathbf{u}\| = 1\}$

If n_1 and n_2 denote the number of elements in S_1 and S_2 , respectively, then find the value of $n_1 - n_2$.

Yes, the answer is correct.

Score: 1

Accepted Answers:

(Type: Numeric) -2

1 point